Repetition, Information Preservation, and Decision Reliability in Hierarchical Language-Model Agents

Anonymous

Anonymous submission draft

Abstract. Repeated reports can help information survive communication, but can also change how a decision maker weighs it. We study these effects in a controlled benchmark for selecting the best feasible candidate schedule and a separate rule-based workflow task. Frozen worker reports distinguish repeated presentation from fresh computation and new evidence. Paired worlds test correct responses to consequential changes and correct stability under harmless edits. A held-out evaluation of Qwen3-14B and Phi-4 uses 192 independent base pairs shared across models, 27,648 decision trajectories, and 55,296 model calls. On scheduling, exact repetition changes accuracy by -0.17 percentage points for Qwen3 and -1.09 for Phi-4, with 95% base-cluster intervals of [-2.21, +1.74] and [-3.17, +1.00]. Corrections and regressions largely offset, and the direct hierarchy interaction is uncertain. An audit of 6,336 development and validation summaries locates numerical objectives becoming Boolean values in raw manager output. A six-pair diagnostic separates missing information from failures to apply a uniquely supported decision. A separate 48-pair mechanism study further tests repetition targets and access to original reports. The results support separate measurement of representation fidelity, decision sufficiency, and final accuracy, while limiting conclusions to the tested tasks, models, and structured interface.

Keywords: language-model agents, repetition, hierarchical aggregation, information preservation, behavioral evaluation

1 Introduction

A manager coordinating language-model agents often receives overlapping reports. Some repeat an observation. Others use similar wording for a different candidate, constraint, or version. The manager must preserve the distinctions that matter, especially when the next decision maker receives only its summary.

This problem arises even when the original reports are correct. In a scheduling team, a worker may report that a plan finishes quickly while another finds that it exceeds machine capacity. Repeating the favorable report adds emphasis without making the plan feasible. Yet repetition could also help a useful number survive a summary boundary. The practical question is when repeated presentation improves the preservation or application of information, and when it distracts from a requirement that changes the decision.

Existing work already studies evidence duplication, source dependence, and hierarchical communication [1, 6, 8]. Distributed-information benchmarks also show that

agent teams can fail to discover important facts [10]. We examine a more restricted setting. All required findings are initially delivered. A flat decision maker reads them directly, while a hierarchical one receives summaries from two intermediate managers. Correct answers follow from executable constraints, allowing us to follow information from its original report to the final decision.

The study asks how repetition affects accuracy, which facts lose their value or scope, when that loss prevents a justified decision, and whether the model applies sufficient information correctly. A missing completion time may be irrelevant if only one candidate is feasible, while a model can choose incorrectly even when all required comparisons are available.

We contribute a controlled candidate-selection benchmark, paired tests of consequential and harmless changes, and a trace-based separation of representation fidelity, decision sufficiency, and final accuracy. The main evaluation covers 192 independent base pairs shared across two models. Its scheduling results do not reproduce the large benefit seen in a small pilot. Corrections and regressions largely offset, and the direct repetition-by-hierarchy interaction is uncertain. A separate numerical audit and completed information-delivery diagnostics show why preserving a fact and applying it correctly require separate evaluation. These findings guide interface assessment without prescribing repetition as a universal remedy or failure source.

2 Related Work

Repeated evidence and fresh computation. Gao et al. [6] study restated observations in controlled logs, web documents, and agent-written communication. Their experiments already include summaries and hierarchical protocols. The agent-team aggregator reads a roster and the full log. Our main root instead receives intermediate summaries and a value-free scope registry, and must combine mandatory constraints with numerical objectives. The distinction concerns complementary information under restricted access, rather than the discovery of evidence double counting. Bara [1] separates report multiplicity from evidence ancestry and correlated extraction errors. These distinctions motivate tracking copies and verification executions separately.

Whole-prompt repetition can improve non-reasoning models in other settings [9]. Our treatment selectively repeats a report within a larger evidence set and may alter what an intermediate manager forwards. It therefore does not directly test or contradict whole-prompt repetition. Self-consistency samples additional reasoning paths before aggregating answers [18]. Multiagent debate exchanges separately generated responses [4], with benefits that depend on protocol and resource comparisons [17]. These methods add computation. Copying an accepted report does not. Likewise, fresh execution on the same source does not establish independent evidence. Work on intrinsic self-correction further distinguishes repeated reasoning from access to external feedback [7].

Organization and distributed information. Kim et al. [8] examine interactions between agent architecture and task structure. We use version 3 of that preprint without importing results from earlier versions. HiddenBench [10] studies failures to explore information held by different agents and evaluates structured communication. Our replay starts

after essential reports have been supplied, isolating later aggregation from information discovery. The MAST taxonomy [2] covers a wider set of coordination and execution failures. Distributed fact recovery under misinformation [19] addresses another boundary. A false report is not the same intervention as a legitimate change in the world. The main replay here contains correct reports in both worlds.

Information fidelity and behavioral evaluation. Summarization research distinguishes source faithfulness from factual correctness [12]. FRANK further separates predicate, entity, and circumstance errors [14]. Our typed scope fields make analogous errors executable to score, while the task supplies an additional criterion of decision sufficiency. CheckList [16] motivates behavioral tests beyond aggregate accuracy, and PAWS [20] shows why lexical overlap does not guarantee equivalent meaning. Position effects [11] and evidence metadata [3] provide alternative explanations for changed answers. They caution against attributing a presentation effect to source counting or a hidden reasoning process without the corresponding intervention.

3 Task Setting and Worked Example

3.1 Selecting a Feasible Candidate

The primary task is to select the supplied feasible schedule with the smallest makespan, the time when its last job finishes. It is not a search for a globally optimal schedule. Jobs have integer durations and resource demands and cannot be interrupted. A job starting at s_j with duration d_j occupies the half-open interval $[s_j, s_j + d_j)$, so finishing at time 2 does not overlap starting at 2. A candidate is feasible only if precedence, release, resource, deadline, and downtime requirements all hold where applicable. Ties follow an explicit public priority order. The answer is `NONE` when no candidate is feasible.

Writing $F(c)$ for the conjunction of mandatory predicates, $M(c)$ for makespan, and $\pi(c)$ for tie priority, the required choice is

$$c^* = \operatorname*{arg\,min}_{c:F(c)=1} \big(M(c), \pi(c)\big). \tag{1}$$

The public decision rule in Eq. 1 first resolves feasibility and then compares objectives. A favorable objective cannot compensate for a failed mandatory constraint.

3.2 Three Jobs, Two Plans, and One Capacity Change

Jobs A and B each need one unit of machine M for two time units. Job C needs one unit of machine N for one time unit and follows both A and B. Machine N has capacity one. Table 1 gives two illustrative schedules. Both independently implemented benchmark checkers verify the example. It is a constructed explanation, not a reported model failure.

At capacity one on M, P fails because A and B overlap. Q is feasible and correct. At capacity two, P becomes feasible and is the correct, faster choice. The objective and precedence reports do not change. The consequential difference is P's resource feasibility.

Table 1. Illustrative schedules. Intervals are half-open and all times use the same unit.

Plan	A on M	B on M	C on N	Makespan
P	$[0,2)$	$[0,2)$	$[2,3)$	3
Q	$[0,2)$	$[2,4)$	$[4,5)$	5

An objective worker contributes the statement that P's makespan is 3. A resource worker contributes the separate finding that P violates M's capacity in the first world. Four presentations of the objective statement still describe one result. They do not provide four independent checks or establish P's overall feasibility.

Now suppose a manager replaces both objective values with `true`. At capacity one, the complete feasibility findings still identify Q as the only possible choice. The objectives are represented incorrectly, but the decision is determined. At capacity two, both candidates are feasible and their missing objective values matter. Finally, if the root selects P after receiving the capacity-one failure and Q's feasibility, it violates sufficient information. This example separates a preserved proposition, enough information to decide, and a correct final decision.

4 Experimental Design

4.1 Instances, Claims, and Paired Worlds

Each base problem has an original and modified world, with every candidate checked by executable rules. The main scheduling set uses six to eight jobs, three candidates, and two or three resource types. An event-sweep checker and an independent integer-time occupancy checker agree on feasibility and objectives. Controlled alterations include fast feasible and infeasible candidates, ties, and cases without a feasible candidate. Names, presentation order, and public tie priority are randomized independently of the answer.

Scheduling mutations change capacities, resource identities, or precedence. A separate workflow task covers quantifiers, conditions, and explicit version applicability using deterministic Boolean requirements and candidate objective values. The task defines version authority explicitly. Its result is secondary. Within each task family, half the main pairs change the correct decision and half preserve it. All variants of one base stay in the same split. Held-out instances reserve graph patterns, parameter combinations, and rendering templates in addition to new seeds. The small structural vocabulary remains a limitation on generalization.

Replay worker reports are generated by deterministic checks and frozen before model inference. They are not new language-model worker executions. A claim binds its value to a task, candidate, entity, predicate, units, condition, and version. Source identifiers, parent reports, and verification execution identifiers record ancestry separately. Similar wording for different candidates or resources is therefore not treated as an equivalent claim. The implementation also separates job precedence, verification tasks, organization, communication access, and report ancestry graphs.

Changed-world pairs regenerate affected reports consistently, so their initial reports are correct. Corrupted reports, autonomous worker errors, and genuine new observations are different interventions. They are not pooled into the main replay. The common record format makes controlled near duplication possible, but a between-world accuracy change alone is not evidence that two within-input claims were falsely merged.

4.2 Presentation, Access, and Aggregation Policy

The baseline displays each required finding once. Exact repetition adds three copies of one selected report. The paraphrase arm adds three fixed-template restatements with the same value, scope, and ancestry. The selected report thus has one or four total presentations. All essential findings remain available. A prespecified rotation counterbalances the target across candidate and task families, preferring a claim unchanged across paired worlds. Selection does not use model outcomes or guarantee a decisive finding.

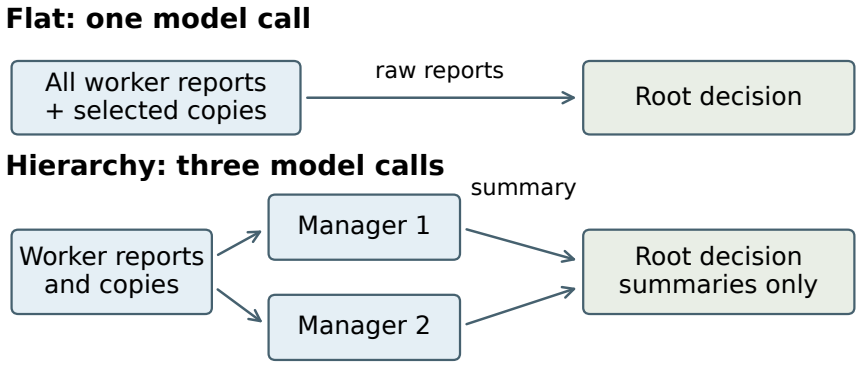


Fig. 1. Flat and hierarchical replay. Copies add presentations, not evidence sources or worker executions.

The access difference in Figure 1 is the central organizational intervention. A flat root reads the full report set in one model call. Two intermediate managers in the hierarchy receive fixed partitions and summarize them for a root, for three calls in total. Copies remain with their original report's branch. The hierarchical root receives summaries and a registry of required scopes and ancestry, without the original assertion values or raw reports. The registry is orchestration metadata, not an oracle answer.

Ordinary synthesis, a required-task checklist, and scope-aware consolidation share the same evidence and decision rule. The checklist explicitly asks for mandatory coverage before optimization. Scope-aware consolidation additionally asks the model to merge equivalent claims, retain differences in entity, condition, or version, and expose unresolved conflicts. It receives no extra truth labels. Because it also adds instructions, an effect cannot be assigned solely to provenance handling.

All policies use a common structured output contract. Managers emit scoped assertions and unresolved keys. Roots emit candidate assessments, a choice, probabilities, and supporting claims. Even ordinary synthesis is thus assisted by a feasibility/objective scaffold. Guided decoding enforces JSON structure but does not require numerical types specifically at objective keys. The analysis scores explicit emitted values and decisions, not private reasoning.

4.3 Models, Evaluation Stages, and Scope

Qwen3-14B [15], with thinking disabled, and Phi-4 [13] run sequentially in bfloat16 without quantization on RTX 6000 Ada Generation GPUs. Weights, tokenizer revisions, native chat templates, and decoding settings are fixed. Sampling uses temperature 0.3, top-p 0.95, a 12,288-token context limit, a 2,304-token output cap, and batches of up to 32. Appendix A records versions and the checkpointed hardware continuation.

Table 2. Reported stages. Bases are shared across models; calls count accepted executions.

Stage	Models	Bases	Sched.	Trajectories	Calls
Pilot	Qwen3	24	12	1,152	2,304
Validation	Both	36	18	5,184	10,368
Main held-out	Both	192	96	27,648	55,296
Mechanism held-out	Qwen3	48	24	576	1,152
Root diagnostic	Qwen3	6 reused	6	192	192

Table 2 distinguishes independent bases from decisions and calls. The main total is $192 \times 2 \times 3 \times 2 \times 3 \times 2 \times 2 = 27{,}648$ trajectories, crossing worlds, presentations, organizations, policies, models, and stochastic generations. Half use one call and half use three, giving 55,296 accepted calls. The mechanism study shares some manager outputs across access arms, so its physical calls differ from the number implied by separate full trajectories.

The pilot has two policies. Validation and main evaluation have three. Matched stage comparisons below retain only synthesis and checklist, while the main headline averages all three. Both models passed the original 85% development competence gate on flat, unrepeated, ordinary-synthesis cases. Scheduling accuracy was 97.92% for Qwen3 and 100% for Phi-4, and workflow accuracy was 87.50% and 93.75%. The gate selected usable models and interfaces before testing. It was not a guarantee of held-out performance or a later exclusion rule. The frozen allocation retained 192 pairs using complete validation results and measured runtime projections, without outcome-based stopping.

The initial questions and analysis were frozen in the repository before main inference. A dated amendment after the pilot introduced the mechanism study, while preserving the original design and without inspecting partial main outcomes. Its separate validation selected 48 fresh pairs under a prospective precision and runtime rule. This is a repository freeze, not an external preregistration. Pilot-inspired trace analyses and the transition analysis introduced for this paper revision are exploratory.

This article reports complete main replay, the original development audits, the completed root diagnostic, and the separate mechanism test. Broader supporting and live-worker comparisons were not completed for both models. The planned objective-only repair replay and separate balanced-objective confirmation are outside this version's evidence. Their absence limits attribution to numerical repair, provenance, position, or worker scaling. No partial two-model supporting results enter the estimates.

4.4 Outcomes, Pairing, and Uncertainty

Exact agreement with the deterministic answer is the primary outcome. A completed malformed or truncated response counts as an operational error, including a failed manager response even if the root happens to choose correctly. Accepted failures are not resampled. Request hashes include the trial and stochastic-generation identity. The completed main archive contains all planned accepted calls, with no unfinished clusters.

For each model, we average accuracy within each base over the relevant worlds, policies, organizations, and generations. The repetition effect is the mean within-base difference

$$\Delta_r = \frac{1}{B}\sum_{b=1}^{B}\left(\bar{Y}_{b,r} - \bar{Y}_{b,\text{single}}\right). \tag{2}$$

Positive values in Eq. 2 mean an accuracy benefit. The original repetition-harm definition uses error and has the opposite sign. The hierarchy interaction directly subtracts the flat repetition effect from the hierarchical effect, averaging exact and paraphrased repetition. Lower hierarchical accuracy alone does not establish amplification.

Paired-world success requires both world answers to be correct. It measures appropriate responsiveness for consequential edits and correct stability for harmless edits. The predeclared contrast compares repetition effects on these two outcomes directly. Two further contrasts compare scope-aware consolidation with the checklist, on accuracy for repeated consequential cases and on retention of changed claims in hierarchical consequential cases. The main scheduling population is primary, and workflow, subgroup, calibration, and transition analyses are secondary or exploratory as identified.

We use 10,000 percentile bootstrap draws of whole base-problem clusters, stratified by mutation family and decision-change status [5]. Each draw retains all worlds, conditions, policies, and generations. Models are reported separately, with pointwise 95% intervals. These are not simultaneous intervals or adjusted significance tests. No confirmatory p-values are reported. The manifest requires Holm adjustment across six primary contrasts if such tests are added. An interval spanning zero does not establish equivalence. The six-base root diagnostic uses counts because its strata cannot support useful intervals.

5 Main Held-Out Results

5.1 Accuracy and Organizational Structure

The scheduling estimates in Table 3 compare identical problems and essential information, with 2,304 trajectories per arm but 96 independent bases. Their small average

effects and uncertainty do not establish selective repetition as a reliable accuracy intervention under this interface.

Table 3. Primary scheduling accuracy and paired changes. Each arm has 96 bases and 2,304 trajectories.

Model	Repetition	Single %	Repeated %	Change, pp [95% CI]
Qwen3-14B	exact	88.72	88.54	-0.17 [-2.21, +1.74]
Qwen3-14B	paraphrase	88.72	88.24	-0.48 [-2.39, +1.35]
Phi-4	exact	80.03	78.95	-1.09 [-3.17, +1.00]
Phi-4	paraphrase	80.03	80.73	+0.69 [-1.09, +2.52]

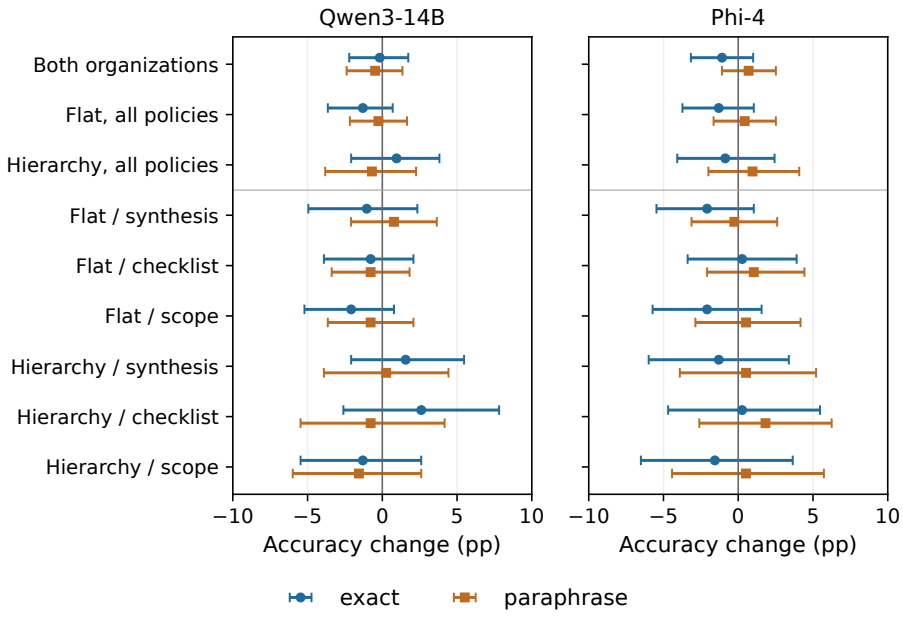


Fig. 2. Scheduling accuracy changes, repeated minus single. Bars are pointwise 95% intervals over 96 bases.

Figure 2 separates pooled effects from organization and policy breakdowns, which vary in direction. Subgroup intervals share the same bases and are not independent replications.

For Qwen3-14B, unrepeated accuracy is 92.97% in the flat organization and 84.46% in the hierarchy. The hierarchical exact-copy effect is +0.95 [-2.08, +3.82] percentage

points. The prespecified hierarchy interaction, averaging exact copies and paraphrases, is +0.91 [-2.17, +3.99] points on the accuracy scale.

For Phi-4, unrepeated accuracy is 88.80% in the flat organization and 71.27% in the hierarchy. The hierarchical exact-copy effect is -0.87 [-4.08, +2.43] percentage points. The prespecified hierarchy interaction, averaging exact copies and paraphrases, is +0.48 [-3.08, +4.17] points on the accuracy scale.

The interaction intervals do not establish that hierarchy consistently changes repetition's effect. Hierarchy changes information access and call count together, so its lower accuracy cannot be assigned to agent count alone.

For Qwen3-14B, scope-aware consolidation minus the checklist changes accuracy on repeated consequential scheduling cases by -0.39 [-3.91, +3.26] points. Its change in retention of altered claims across all hierarchical consequential cases is -0.78 [-3.39, +1.22] points. Each comparison uses 48 independent bases.

For Phi-4, scope-aware consolidation minus the checklist changes accuracy on repeated consequential scheduling cases by -1.30 [-5.34, +2.86] points. Its change in retention of altered claims across all hierarchical consequential cases is -3.12 [-5.27, -1.13] points. Each comparison uses 48 independent bases.

These policy contrasts give no clear accuracy advantage to the richer consolidation instructions over the checklist. The Phi-4 retention estimate is lower under consolidation in this comparison, a reminder that more detailed instructions need not preserve more information. The common registry and output contract already assist all policies. Appendix B provides the absolute cell accuracies needed to judge these comparisons.

5.2 Consequential Changes and Harmless Controls

Table 4 requires correctness before and after the world edit. A correct answer in just one world is a failure on this stricter outcome. The exact-copy pattern differs across models and edit classes, and the predeclared difference of effects uses both repetition forms.

Table 4. Both worlds correct (%), scheduling. Each row has 48 bases and 576 paired cells per arm.

Model	Edit	Single	Exact	Para.	Exact change, pp [95% CI]
Qwen3-14B	harmless	80.03	78.12	78.12	-1.91 [-6.08, +1.91]
Qwen3-14B	consequential	83.85	88.37	86.63	+4.51 [+0.52, +8.51]
Phi-4	harmless	68.75	71.01	71.01	+2.26 [-1.56, +6.08]
Phi-4	consequential	72.57	70.14	73.26	-2.43 [-6.25, +1.39]

For Qwen3-14B, the direct consequential-minus-harmless difference in the repetition effect on paired success is +5.56 [+0.09, +11.20] points, averaging exact and paraphrased repetition. For Phi-4, the direct consequential-minus-harmless difference in the repetition effect on paired success is -3.12 [-7.90, +1.74] points, averaging exact and paraphrased repetition.

Qwen3's contrast favors consequential pairs, whereas Phi-4's interval includes either direction. This is neither a shared tendency to ignore changes nor evidence of equivalent sensitivity. The test concerns response to a world edit. Identifying loss of an exception within one input requires a separate claim-level test.

5.3 Corrections, Regressions, and Stochastic Variation

The near-zero mean effects can hide substantial turnover. Table 5 distinguishes a wrong answer becoming correct, a correct answer becoming wrong, and a change between two wrong answers. For example, Qwen3's 93 corrections and 97 regressions give a net loss of four correct decisions out of 2,304, explaining its small mean effect. The answer-change count is larger because some wrong answers change to other wrong answers.

Table 5. Scheduling transitions. Answer changes require two valid trajectories; counts are paired comparisons.

Model	Comparison	N	Corrected	Spoiled	Wrong/wrong	Answer changes
Qwen3-14B	Single, 0 to 1	1152	28	28	7	62/1151
Qwen3-14B	Exact, 0 to 1	1152	31	25	4	60/1152
Qwen3-14B	Single to exact	2304	93	97	24	213/2303
Phi-4	Single, 0 to 1	1152	70	66	15	148/1149
Phi-4	Exact, 0 to 1	1152	97	60	22	177/1149
Phi-4	Single to exact	2304	143	168	51	357/2298

Qwen3-14B has 93 corrections and 97 regressions under exact repetition. Its within-condition answer-change rate is 5.38% without repetition and 5.21% with exact repetition. The latter minus the former is -0.17 [-1.74, +1.39] percentage points.

Phi-4 has 143 corrections and 168 regressions under exact repetition. Its within-condition answer-change rate is 12.85% without repetition and 15.36% with exact repetition. The latter minus the former is +2.52 [+0.61, +4.51] percentage points.

The first two comparisons in Figure 3 use generation 0 versus 1 within a condition, giving 1,152 pairs per model. The third compares no repetition with exact repetition at the same generation index and retains both indices, giving 2,304 pairs. Seeds are shared across conditions at a given base, index, and stage. Different-seed within-condition variation and common-seed cross-condition variation therefore use different random couplings. The larger cross-condition count cannot by itself establish that repetition increases stochastic instability.

Comparing within-condition disagreement between treatments is closer to that question. Qwen3's estimate is near zero and uncertain. Phi-4's exact-copy arm has more answer disagreement across its two generations in this exploratory analysis. There is still only one different-seed pair per world, policy, and organization. Reported answer-change rates use all paired cells, count changes only when both trajectories are valid, and list the valid-answer denominator separately. These limits prevent converting a descriptive transition pattern into a universal claim about stability.

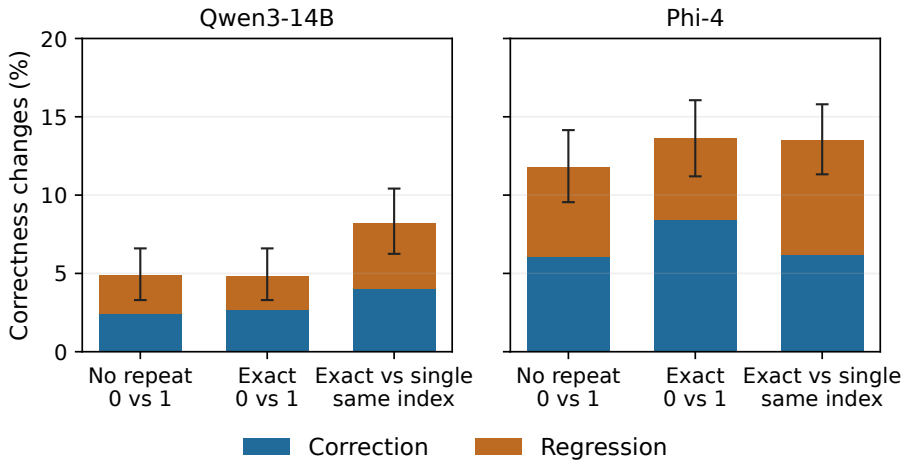


Fig. 3. Corrections and regressions in scheduling. Bars show 95% base-cluster intervals for total correctness changes.

5.4 Generalization Beyond the Pilot

Under the shared synthesis/checklist policies, the hierarchical Qwen3 exact-copy effect is +13.54 [+4.17, +22.92] points in the 12-base pilot, -3.47 [-10.42, +2.78] in the 18-base validation subset, and +2.08 [-1.56, +5.60] in the 96-base main subset. These averages retain both worlds and both stochastic generations. The main two-policy estimate therefore differs from the three-policy headline. One pilot base contributes seven of 14 corrections. It remains included in every primary estimate. The pilot pattern is less pronounced in the later evaluations, which also change graph patterns, parameter combinations, and language templates. This establishes a limit on generalization of the development finding, without identifying which changed feature explains it.

Figure 6 in the appendix displays every main scheduling base in a fixed identifier order. Positive and negative effects occur across problems, without selecting exclusions or a favorable subpopulation. The main inference continues to weight each independent base equally.

5.5 Secondary Workflow Findings

Table 6 shows a decrease for Phi-4 under both repetition forms, while Qwen3's estimates remain close to zero with uncertain direction. The workflow shares the feasibility-before-objective structure but uses different rule types. It is a separate secondary population, not another scheduling replication.

The original development gate passed for both models. On held-out flat, unrepeated, ordinary-synthesis workflow cases, accuracy is 84.38% for Qwen3 and 83.33% for Phi-4, below the same 85% reference. The frozen protocol retains these cases and reports the decline as a limitation on secondary interpretation. It does not remove a difficult task or

Table 6. Secondary workflow accuracy. Each arm has 96 bases and 2,304 trajectories.

Model	Repetition	Single %	Repeated %	Change, pp [95% CI]
Qwen3-14B	exact	78.69	78.91	+0.22 [-2.17, +2.43]
Qwen3-14B	paraphrase	78.69	78.52	-0.17 [-2.26, +1.82]
Phi-4	exact	74.65	71.66	-2.99 [-5.03, -1.09]
Phi-4	paraphrase	74.65	71.79	-2.86 [-5.03, -0.78]

retune the model after testing. The workflow intervals remain pointwise and secondary, so a decrease should not be promoted to a family-wide confirmatory significance claim.

6 Information Preservation and Decision Sufficiency

6.1 Where Numerical Information Changes

The pipeline audit follows original typed reports, manager inputs, raw generated summaries, parsed summaries, and actual root inputs. It covers 60 independent pilot and validation bases, 3,168 hierarchical trajectories, and 6,336 manager summaries. These are different denominators. No field changes were found between the checked raw manager output, parser output, and root delivery. Numerical objectives becoming Boolean `true` were already present in raw model responses.

The prompts require numerical objectives, but a common assertion schema permits both Boolean and numerical values without conditioning the value type on the scope key. A syntactically valid response can therefore lose the objective's meaning. The audit locates the error in generated communication under this interface. It does not establish whether a different schema, instruction example, or model would eliminate it.

Table 7. Objective-field audit. Entries give single / exact counts; field denominators are per arm.

Scheduling subset	Fields	Preserved	Boolean	Omitted
Pilot Qwen3	288	219 / 270	60 / 9	9 / 9
Validation Qwen3	648	561 / 549	53 / 64	34 / 35
Validation Phi-4	648	458 / 449	156 / 156	25 / 40

Table 7 and Figure 4 show why the large pilot improvement should not be interpreted as a general preservation mechanism. Boolean replacement falls markedly with exact copies in the pilot, but the later model and presentation cells have different patterns. The pilot averages two policies, and validation averages three, so this display describes their audited fields rather than isolating a stage intervention. A field is preserved only when its correct value is emitted at the correct scope without an unresolved conflict. A correct number attached to another candidate does not qualify. The displayed categories are not exhaustive or mutually exclusive. A correct value marked unresolved does not count as preserved.

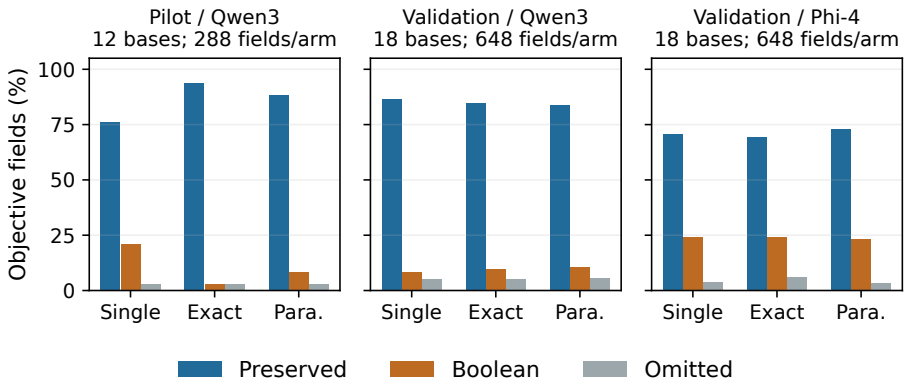


Fig. 4. Objective fidelity in the development audit. Counts use distinct scoped fields, not repeated copies.

The historically influential pilot base illustrates the transformation. All eight baseline objective-bearing summaries replace numerical objectives with Boolean values, while all eight corresponding exact-copy summaries preserve the numbers. The repeated report concerns a candidate other than the winner. This trace was selected by its largest contribution to pilot corrections and is interpreted alongside aggregate rates and intact controls, not as a representative frequency. The reproducibility record retains a fixed identifier-ordered selection of Boolean-loss, omission, altered-value, and intact traces where available.

6.2 A Fact Can Be Missing Without Making the Choice Ambiguous

Representation fidelity asks whether a proposition retains its source value and scope. Decision sufficiency asks whether all admissible completions of unknown fields lead to the same answer. Final accuracy asks whether the model selects the correct candidate in the actual world. None substitutes for the others.

The scheduling support scorer operates on the public report interface. Mandatory predicates are Boolean and objectives are nonnegative integers. It holds supplied, uncontested values fixed and enumerates possible feasibility combinations and objective orderings, including ties. A unique resulting answer is determined by the visible information. Multiple answers mean the input does not uniquely justify a choice. The scorer does not fill gaps with evaluator truth or infer unseen schedule details from hidden constraints.

Contradictory and ill-typed fields are marked separately. The strict representation has no consistent completion if it contains such fields. A complementary support analysis treats only those disputed or invalid fields as unknown, while retaining uncontested values. It can establish that the choice is unchanged under every admissible replacement, as in the sole-feasible-candidate example. This does not make the invalid representation correct or resolve the disputed facts.

Table 8. Decision support from uncontested summary fields. Representation-error cases remain flagged.

Scheduling subset	Trajectories	Unique choice	With invalid fields	Violations
Pilot Qwen3	288	248	8	4
Validation Qwen3	648	585	40	73
Validation Phi-4	648	411	39	52

Table 8 counts this distinction across all three presentation conditions. Its invalid-field column is a subset of inputs with a determined choice. The violations column counts valid final choices that disagree with that choice, rather than merely checking world accuracy. The pilot uses two policies and validation uses three. Conditioning on observed information preservation is descriptive, since repetition can itself change that property. These rates are not a causal mediation estimate.

6.3 Fixed-Summary Root Replay and Lossless Delivery

The root diagnostic uses six reused development scheduling pairs, both worlds, and one fresh root call per arm. Its 192 calls cross eight root-level treatments with two delivery modes. Ordinary synthesis and response requirements are fixed. Within a delivery mode, the incoming distinct propositions remain fixed across repetition conditions. The arms repeat one candidate objective, all candidate objectives, or a constraint already present, alongside neutral and position controls. There is no paraphrase arm.

The second delivery mode deterministically carries the original worker assertions available to each branch, preserving values and scopes. It supplies no hidden answer or new observation. It may restore information lost in model summaries, so it is a different information-delivery intervention, not repetition of an identical input. The no-repetition root response is freshly generated in each mode. Manager calls are historical or deterministic and are not included in the 192 new root calls.

Without root-level repetition, frozen model summaries yield 8 correct decisions out of 12 and lossless delivery yields 12 out of 12. Of the frozen contexts, 10 determine a unique choice from uncontested values, including 2 with irrelevant invalid objective fields. The root violates that unique choice in 2 cases. The other 2 contexts remain underdetermined. Mean root input length is 3,333 tokens for frozen summaries and 3,333 for lossless delivery. The latter changes the represented information and its presentation, so the comparison is an idealized preservation intervention. It does not isolate the effect of repairing only numerical objectives.

Repeating all objective assertions already present in the frozen summaries yields 12 correct world decisions out of 12. However, only 10 are uniquely supported by that input. The other 2 correct choices remain unsupported by a unique visible answer. Uniform repetition also repeats any existing Boolean objective values rather than replacing them with hidden numbers. With lossless delivery, the unrepeated baseline and uniform repetition both yield 12 out of 12. Individual-objective and neutral-material arms include regressions, as reported in Appendix C. These six reused development bases

support a diagnosis of observable failure locations, not a precise population effect or a complete mediation analysis.

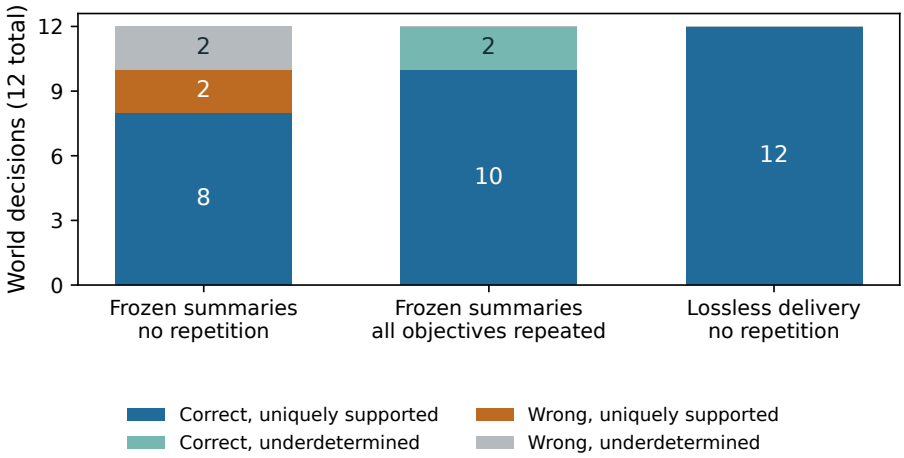


Fig. 5. Root diagnostic on 12 worlds from six bases. World correctness and uniquely supported correctness differ.

Figure 5 distinguishes a correct world answer from enough information to justify it. The interventions have the same observed accuracy but differ in whether all decisions are determined. Neither reveals a hidden reasoning path.

6.4 A Separate Test of Repetition Targets and Raw-Report Access

The Qwen3 mechanism study uses 48 fresh pairs, split equally between scheduling and workflow, with 576 trajectories. A decisive report is a correct mandatory finding whose isolated truth-value flip changes the complete-information answer. A common target is a true favorable finding for an infeasible candidate when available, otherwise a correct nondecisive Boolean finding. These categories come from executable eligibility rules before inference, not from model outcomes. The selected scope may differ between the two worlds.

The six arms compare no repetition, selective repetition of a common or decisive finding, uniform repetition of all required reports, and access to both summaries and raw reports with or without common repetition. Raw-access arms reuse the same manager summaries as their corresponding hierarchy arm. All evidence was already available to the managers. Necessary-claim retention requires the correct scoped value to survive without a contradictory value or unresolved flag. A claim is necessary when removing it makes the decision undetermined.

The primary decisive-minus-common accuracy contrast is -2.08 [-10.42, +6.25] points. Repeating a common finding changes retention of unrepeated necessary claims

Table 9. Separate Qwen3 mechanism test. Each task has 24 fresh bases and 48 decisions per arm.

Arm	Schedule %	Workflow %	Sched. retention %
No repetition	87.50	83.33	95.83
Common finding, four	87.50	75.00	95.57
Decisive finding, four	85.42	91.67	99.11
All reports, four each	79.17	70.83	96.09
Raw access, no repetition	85.42	83.33	95.83
Raw access, common four	81.25	72.92	95.57

by -0.26 [-0.78, 0.00] points. The repetition-by-raw-access interaction is +4.17 [-4.17, +10.42] points. Neither accuracy contrast establishes a general benefit from emphasizing decisive findings or from bypassing summaries. The retention estimate is small and does not demonstrate equivalence.

On the secondary workflow population, uniform repetition lowers accuracy from 83.33% to 70.83%, a change of -12.50 [-20.83, -4.17] points. This is a secondary comparison in a single-model extension. Repeating every report also substantially lengthens the input, and these arms do not isolate exceptions or context length. The result therefore cannot establish general harm to exception handling.

In Table 9, scheduling accuracy is identical for baseline and common repetition, while target and access contrasts remain uncertain. Retention averages the per-trajectory fraction of eligible unrepeated necessary scopes, including missing values as losses. The prespecified primary contrasts are target specificity, unrepeated necessary-claim retention, and the repetition-by-access interaction. Other arm comparisons and all workflow effects are secondary. A retention increase and an accuracy change need not coincide.

7 Discussion and Limitations

7.1 Implications for Evaluating Agent Communication

The main results favor a more specific evaluation question than whether repetition helps. An average can remain almost unchanged while some previously correct decisions become wrong and others are corrected. Evaluating those transitions alongside ordinary stochastic variation makes the distinction visible without equating every answer change with a new instability mechanism.

The communication audit suggests an equally specific engineering target. A structured output can be syntactically valid while carrying the wrong kind of value. Scope-dependent type validation is a plausible way to prevent a Boolean from being accepted at an objective key, but its effect on final decisions has not been tested here. An interface improvement should be assessed for preserved propositions, sufficient information, and correct application separately.

The root diagnostic shows why these measures should remain distinct. Some lost objectives do not affect a uniquely determined choice, and some correct choices are not justified by the delivered information. A system that is right for an unsupported reason can fail when the same missing distinction becomes relevant. Conversely, a faithful

summary does not by itself guarantee that the root will apply a mandatory constraint or choose the best feasible candidate.

7.2 Boundaries of the Conclusions

The benchmark is synthetic and deliberately controlled. It measures integration of complementary requirements rather than deployment-wide failure rates. The workflow shares the selection structure with scheduling, limiting domain-transfer claims. Frozen correct reports isolate aggregation but omit the error process of autonomous workers. A small fresh-verification diagnostic is summarized in the appendix and does not establish independence of worker errors.

Organization changes both input access and the number of model calls. The main result therefore concerns the tested organizational package. Incomplete broader call-matched, length, ancestry, and live-worker controls prevent stronger claims about those components. The completed raw-access comparison helps probe the summary boundary but also changes root input length. Uniform repetition increases length more than selective repetition. Neither contrast alone identifies why a decision changed.

Both models use one fixed precision, decoding regime, and structured interface. The common candidate-assessment scaffold narrows the contrast between ordinary synthesis and the checklist. The scope policy changes instructions as well as consolidation guidance. Generalization to other model sizes, reasoning modes, free-form communication, deeper organizations, or feedback loops remains open.

The original main contrasts were frozen before inference, but pilot-inspired audits and the stochastic-transition analysis are exploratory. Pointwise intervals across multiple related comparisons are not family-wise significance tests. The six-pair diagnostic and the 24 scheduling pairs in the mechanism extension have limited precision. No equivalence margin was prospectively specified for the fresh results. Trace associations locate observable changes but do not identify causal mediation or a faithful internal reasoning process.

8 Conclusion

Across two models and 192 independent held-out base pairs, selective repetition has small, uncertain average scheduling effects and does not reproduce the large hierarchical pilot gain. Corrections and regressions can offset, so average accuracy alone does not describe how decisions change. The secondary workflow results and targeted repetition study further limit any claim of a universal benefit. Trace audits identify numerical-to-Boolean transformations in raw summaries, while the root diagnostic separates lost information from errors made despite a uniquely supported choice. These findings support evaluating the complete communication path through three distinct outcomes: faithful propositions, sufficient information, and correct decisions. They motivate interfaces that preserve required values and scopes, with their decision benefit assessed separately.

A Reproducibility and Output Semantics

The accompanying materials contain generators, independent verifiers, templates, frozen configurations, immutable requests and responses, and analysis scripts. All reported tables and figures are generated from complete local records. Identifying repository links are omitted for anonymous review. Original analyses are preserved alongside an analysis-only correction that sorts cluster identifiers before bootstrap resampling. That correction changes Monte Carlo endpoints for the hierarchy interaction, not the decisions, estimator, strata, or experiment. The bootstrap seed is 20271015. The frozen model-averaging rule gives each model equal weight, while this paper presents model-specific estimates.

Model and tokenizer revisions are identical within each family. Qwen3-14B uses

```
40c069824f4251a91eefaf281ebe4c544efd3e18
```

and Phi-4 uses

```
2db69c1c3e91a05d2c64a3185acfbaf36f744e25
```

Their licenses are Apache 2.0 and MIT. The environment uses vLLM 0.10.2, PyTorch 2.8.0 with CUDA 12.8, and Transformers 4.55.4. Checkpointed continuation used RTX 6000 Ada Generation hardware with 49,140 MiB memory. Recorded driver versions are 595.91.07 and 570.124.06 across the original and replacement allocations. Frozen model, prompt, precision, and sampling settings were retained. Prefix caching is disabled. Native chat-template token IDs are submitted directly, with template dates pinned to 16 October 2026. Seeds derive from base identifier, generation index, and stage. Distinct generation indices retain distinct requests.

Manager assertions are triples of scope key, value, and cited report identifiers, for example `["objective|P|...", 3, ["report-id"]]`. The surrounding JSON also lists unresolved keys. Roots add candidate assessments, a choice, and probabilities. The scope registry carries units and applicability. A Boolean is never scored as a valid numerical objective. Integral floats such as 3.0 are numerically equivalent to 3. Correctness includes operational failures as errors. Invalid selections use all trajectories as the denominator, while calibration requires a valid probability distribution and reports its available denominator in the analysis files.

B Policy Breakdown and Resource Use

Table 10 provides absolute scheduling accuracy for each policy. These values explain why the pooled comparison should not be attributed to provenance instructions alone. Scope-aware consolidation has no consistent accuracy advantage across model and organization cells.

Table 11 counts input and output tokens, including the managers in a hierarchical trajectory. Exact repetition adds processing even when average accuracy does not improve. Allocated time is measured inference batch wall time shared among requests, summed over the trajectory and divided by correct answers. It excludes loading and

Table 10. Main scheduling accuracy (%) by policy. Each cell has 96 bases and 384 trajectories.

Model	Organization	Policy	Single	Exact	Para.
Qwen3-14B	flat	synthesis	91.15	90.10	91.93
Qwen3-14B	flat	checklist	95.05	94.27	94.27
Qwen3-14B	flat	scope	92.71	90.62	91.93
Qwen3-14B	hierarchy	synthesis	87.76	89.32	88.02
Qwen3-14B	hierarchy	checklist	78.65	81.25	77.86
Qwen3-14B	hierarchy	scope	86.98	85.68	85.42
Phi-4	flat	synthesis	94.01	91.93	93.75
Phi-4	flat	checklist	89.06	89.32	90.10
Phi-4	flat	scope	83.33	81.25	83.85
Phi-4	hierarchy	synthesis	80.21	78.91	80.73
Phi-4	hierarchy	checklist	63.28	63.54	65.10
Phi-4	hierarchy	scope	70.31	68.75	70.83

Table 11. Scheduling resources, single / exact. Time is allocated batch seconds per correct decision.

Model	Organization	Single tokens	Exact tokens	Single time	Exact time
Qwen3-14B	flat	4,230	4,526	2.45	2.51
Qwen3-14B	hierarchy	9,071	9,602	10.46	10.58
Phi-4	flat	3,709	3,967	2.14	2.26
Phi-4	hierarchy	8,001	8,455	10.21	10.61

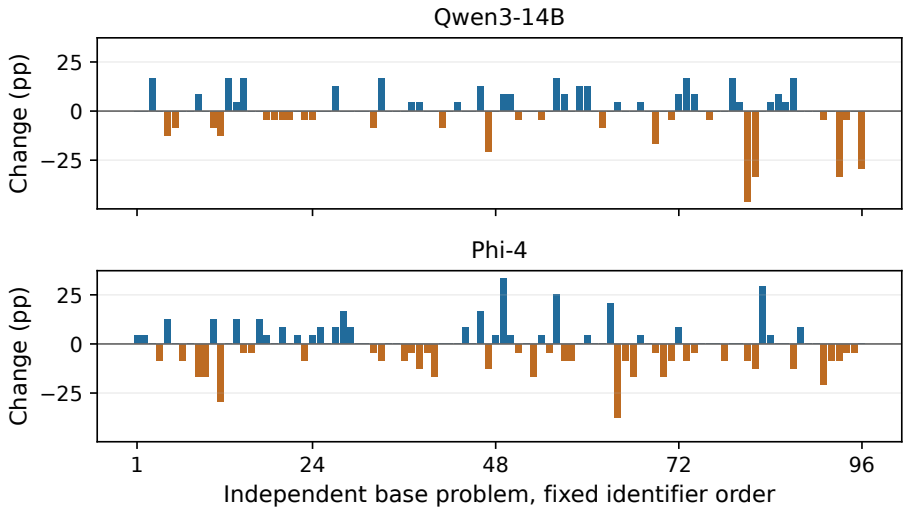


Fig. 6. Exact-copy scheduling effects for all 96 bases. Each bar averages organizations, policies, worlds, and generations.

setup, and is neither individual request latency nor measured monetary cost. Shared manager calls in mechanism studies are counted once in their physical inventories.

Figure 6 shows how a small mean can conceal changes on individual problems. Fixed identifier ordering includes every base. These descriptive effects do not define new confirmation subsets.

C Complete Root-Diagnostic Arms and Fresh Verification

Table 12 reports every root-diagnostic arm, with twelve world decisions from the same six bases. Objectives 1 to 3 follow the prespecified candidate ordering. Agreement is counted only where uncontested facts determine a unique answer. That denominator remains ten under frozen summaries and twelve under lossless delivery. Neutral material changes no task evidence. The end arm moves copies of the designated objective assertion to the end.

Table 12. All root-diagnostic arms. Agreement uses only uniquely determined contexts.

Arm	Frozen correct	Agreement	Lossless correct	Agreement
No repetition	8/12	8/10	12/12	12/12
Objective 1	9/12	9/10	11/12	11/12
Objective 2	9/12	9/10	12/12	12/12
Objective 3	9/12	9/10	12/12	12/12
Constraint	9/12	9/10	12/12	12/12
All objectives	12/12	10/10	12/12	12/12
Neutral material	9/12	9/10	11/12	11/12
Objective at end	8/12	8/10	11/12	11/12

The lossless baseline's observed ceiling does not imply that its surrounding prompt is immune to presentation changes. Individual-objective, position, and neutral arms each include a wrong decision in at least one lossless context. Uniform repetition under frozen summaries reaches twelve correct world answers while two contexts remain underdetermined. The complete table prevents selecting only the largest gain as evidence for a general repetition effect.

A separate six-base development verification probe completed 30 trajectories and 72 physical calls. Original reports, exact copies, paraphrases, added fresh verification, and a fixed-allowance fresh variant all yielded six correct final decisions out of six. Fresh checks corrected one initial worker error and also introduced an incorrect finding. Roots had authoritative task data in every arm, so their ceiling performance cannot be attributed solely to worker reports. These calls share evidence sources and do not mea-

sure independent observations or verified computational diversity. Their small sample supports a diagnostic comparison rather than a worker-scaling conclusion.

Declaration on AI Assistance

The author supplied the research direction and experimental requirements. An AI coding assistant assisted with protocol implementation and methodological refinements, software development and testing, experiment operation, literature verification, statistical analysis, figure generation, and manuscript drafting. This draft has not yet undergone the author's scientific review. The author must critically verify and revise the scientific arguments, analyses, references, and conclusions, and assess the actual contributions against the conference policy before submission. The assistance was not limited to language editing. No AI system is listed as an author.

References

1. Bara, M.: Epistemic Sybil Resistance: Multiplying AI Agents Without Multiplying Evidence. arXiv:2609.01873v1 (2026). Preprint. https://arxiv.org/abs/2609.01873v1
2. Cemri, M., Pan, M.Z., Yang, S., Agrawal, L.A., Chopra, B., Tiwari, R., et al.: Why Do Multi-Agent LLM Systems Fail? Advances in Neural Information Processing Systems **38**, Datasets and Benchmarks Track (2025). https://papers.nips.cc/paper_files/paper/2025/hash/b1041e52d3be19f0a9bc491657488e4a-Abstract-Datasets_and_Benchmarks_Track.html
3. Chiang, C.H., Lee, H.: Do Metadata and Appearance of the Retrieved Webpages Affect LLM's Reasoning in Retrieval-Augmented Generation? In: Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP, pp. 389–406 (2024). https://doi.org/10.18653/v1/2024.blackboxnlp-1.24
4. Du, Y., Li, S., Torralba, A., Tenenbaum, J.B., Mordatch, I.: Improving Factuality and Reasoning in Language Models through Multiagent Debate. In: Proceedings of the 41st International Conference on Machine Learning, PMLR 235, pp. 11733–11763 (2024). https://proceedings.mlr.press/v235/du24e.html
5. Field, C.A., Welsh, A.H.: Bootstrapping clustered data. Journal of the Royal Statistical Society: Series B (Statistical Methodology) **69**(3), 369–390 (2007). https://doi.org/10.1111/j.1467-9868.2007.00593.x
6. Gao, J., Li, R., Yu, T., Ren, L., Chen, B., Wang, Z.: Copies or Sources? Measuring How LLM Aggregators Count Restated Evidence in Multi-Agent Systems. arXiv:2610.06192v1 (2026). Preprint. https://arxiv.org/abs/2610.06192v1
7. Huang, J., Chen, X., Mishra, S., Zheng, H.S., Yu, A.W., Song, X., et al.: Large Language Models Cannot Self-Correct Reasoning Yet. In: International Conference on Learning Representations (2024). https://proceedings.iclr.cc/paper_files/paper/2024/hash/8b4add8b0aa8749d80a34ca5d941c355-Abstract-Conference.html
8. Kim, Y., Gu, K., Park, C., Park, C., Schmidgall, S., Heydari, A.A., et al.: Towards a Science of Scaling Agent Systems. arXiv:2512.08296v3 (2026). Preprint, revised 8 April 2026. https://arxiv.org/abs/2512.08296v3
9. Leviathan, Y., Kalman, M., Matias, Y.: Prompt Repetition Improves Non-Reasoning LLMs. arXiv:2512.14982v1 (2025). Preprint. https://arxiv.org/abs/2512.14982v1
10. Li, Y., Naito, A., Shirado, H.: Systematic Failures in Collective Reasoning under Distributed Information in Multi-Agent LLMs. In: Proceedings of the 43rd International Conference on Machine Learning, PMLR 306, pp. 69250–69268 (2026). https://proceedings.mlr.press/v306/li26ej.html

11. Liu, N.F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F., et al.: Lost in the Middle: How Language Models Use Long Contexts. Transactions of the Association for Computational Linguistics **12**, 157–173 (2024). https://doi.org/10.1162/tacl_a_00638
12. Maynez, J., Narayan, S., Bohnet, B., McDonald, R.: On Faithfulness and Factuality in Abstractive Summarization. In: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, pp. 1906–1919 (2020). https://doi.org/10.18653/v1/2020.acl-main.173
13. Microsoft: Phi-4. Official model card, pinned revision listed in Appendix A. Accessed 18 October 2026. https://huggingface.co/microsoft/phi-4
14. Pagnoni, A., Balachandran, V., Tsvetkov, Y.: Understanding Factuality in Abstractive Summarization with FRANK: A Benchmark for Factuality Metrics. In: Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, pp. 4812–4829 (2021). https://doi.org/10.18653/v1/2021.naacl-main.383
15. Qwen Team: Qwen3-14B. Official model card, pinned revision listed in Appendix A. Accessed 18 October 2026. https://huggingface.co/Qwen/Qwen3-14B
16. Ribeiro, M.T., Wu, T., Guestrin, C., Singh, S.: Beyond Accuracy: Behavioral Testing of NLP Models with CheckList. In: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, pp. 4902–4912 (2020). https://doi.org/10.18653/v1/2020.acl-main.442
17. Smit, A.P., Grinsztajn, N., Duckworth, P., Barrett, T.D., Pretorius, A.: Should we be going MAD? A Look at Multi-Agent Debate Strategies for LLMs. In: Proceedings of the 41st International Conference on Machine Learning, PMLR 235, pp. 45883–45905 (2024). https://proceedings.mlr.press/v235/smit24a.html
18. Wang, X., Wei, J., Schuurmans, D., Le, Q., Chi, E.H., Narang, S., et al.: Self-Consistency Improves Chain of Thought Reasoning in Language Models. In: International Conference on Learning Representations (2023). Camera-ready version: https://arxiv.org/abs/2203.11171v4
19. Yan, C., Yue, Z., Zhao, F., Lin, E., Jia, L., Tong, H., et al.: When Truth Is Distributed: Misinformation Derails Collective Fact Recovery in LLM-Based Multi-Agent Systems. arXiv:2608.03421v1 (2026). Preprint. https://arxiv.org/abs/2608.03421v1
20. Zhang, Y., Baldridge, J., He, L.: PAWS: Paraphrase Adversaries from Word Scrambling. In: Proceedings of NAACL-HLT, Volume 1, pp. 1298–1308 (2019). https://doi.org/10.18653/v1/N19-1131